Multiply Two matrices :

```
a1 =[int(x) for x in input().split() ]
a2 =[int(x) for x in input().split() ]
a3 =[int(x) for x in input().split() ]
b1 =[int(x) for x in input().split() ]
b2 =[int(x) for x in input().split() ]
b3 =[int(x) for x in input().split() ]

A= [a1, a2, a3]
B= [b1, b2, b3]

R = []
for i in range(len(A)):
    toAdd = len(A[0])*[0]
    R.append(toAdd)
##R initialized with all zeros

for row in range(len(A)):
    for col in range(len(A[0])):
        for aux_pos in range(len(A)):
            R[row][col] += A[row][aux_pos]*B[aux_pos][col]



for row in R:
    print(*row)
```

Find leap year :

```
s=input()
d = int(s[-4:])
lst = []
while(len(lst) < 50):
    if((d % 100 != 0 and d % 4 == 0) or (d % 100 == 0 and d % 400 == 0)):
        lst.append(d)
        d += 4
    else:
        d = d + (4 - (d % 4))

print(lst)
```

Sam is playing with a list :

```
arr = [int(i) for i in input().split()]
k = int(input())

for i in range(len(arr)-k+1):
    print(len(set(arr[i: i+k])), end=' ')
```

Mix it :

```
s = input().split()
s1, s2 = s
final = ""
if (len(s1) > len(s2)) :
    for i in range(len(s2)) :
        final += s1[i]
        final += s2[i]
    final += s1[len(s2):]
else :
    for i in range(len(s1)) :
        final += s1[i]
        final += s2[i]
    final += s2[len(s1):]
print(final)
```

Character change :

```
s=input()
x=input()
c=input()
c1=0
for i in range(len(s)-len(x)+1):
    t=s[i:i+len(x)]
    f=0
    for j in range(len(x)):
        if x[j]==c:
            continue
        else:
```

```
            if x[j]!=t[j]:
                f=1
        if f==0:
            c1+=1
print(c1)
```

Leader :

```
L1 = [int(x) for x in input().split()]


L2 = [max(L1[:i]) for i in range(1, len(L1) + 1)]

print(*L2)
```

Help Jane :

```
def checkSimilar(x,y):
    if len(x) != len(y):
        return False
    dic = {}
    for i in range(len(x)):
        xx = x[i]
        yy = y[i]
        if(xx in dic.keys()):
            if(dic[xx] != yy):
                return False
        else :
            dic[xx] = yy
    return True

if __name__ =="__main__":
    x = input()
    y = input()
    if checkSimilar(x,y):
```

```
    print("YES")
else:
     print("NO")
```